

AUSSEF

Complete experiment record

A concise record of every distinct feasibility audit, measurement study, statistical test and predictive experiment completed across the project.

Prepared for discussion with Bowen

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Executive summary

AUSSEF has tested the same central proposition through several levels of evidence: whether observed disaster exposure, local vulnerability, fiscal condition, network structure or recovery funding can predict measurable socioeconomic or infrastructure consequences. The project has produced a strong audit trail and several measurable descriptive effects, but no model has yet shown a repeatable advantage across genuinely unseen disaster groups for the intended substantive outcome.

23	distinct completed experiments or formal decision gates
1	stable event-time functional response: the 2016 South Australian blackout demand decline
0	validated general predictors of disaster socioeconomic impact or multidimensional infrastructure criticality

What has been learned

The main constraint is not algorithm choice. It is target validity, temporal independence and event-level coverage. Large row counts repeatedly collapse to a handful of independent seasons, disasters, council-events or infrastructure failures. Annual income and fiscal ratios are observable, but they are indirect and heavily confounded. More flexible models fit historical rows yet fail forward or disaster-family validation.

The strongest positive contribution is methodological: the project now distinguishes source rows from independent events, missing values from zeros, functional disruption from economic harm, gross burden from funding receipts, and descriptive association from causal or predictive evidence.

Positive and suggestive findings retained

1. South Australian electricity demand fell 26.0% after the 2016 blackout in the common response protocol, with the same direction across all tested bandwidths. This is a functional-disruption result.
2. A five-case fiscal recovery sensitivity gave M2 ridge MAE of 5.805 percentage points, 0.529 better than M0, but the result disappeared in the seven-case cohort and did not reproduce for maintenance outcomes.
3. Naughtons Gap and Fitzroy appear conventionally underrated in a four-case descriptive criticality comparison. The sample is too small for inference.
4. The bushfire interaction estimates and some feature importances are real fitted associations, but signs and magnitudes do not remain stable across seasons, outcomes and sensitivity designs.

Overall research position

MEASUREMENT PROGRAM: SUCCESSFUL | GENERAL PREDICTIVE CLAIM: NOT ESTABLISHED

The evidence supports presenting AUSSEF as a rigorous sequence of attempted falsifications and measurement improvements. It does not support selecting a preferred model from repeated negative trials or describing missing outcome chains as zero disaster effects.

Master experiment register

#	Experiment	Primary test	Decision
1	Four-dimension data feasibility	DL / IL / FP / SL construction	GO for bounded construction; full index NO-GO
2	Six-case bushfire pilot	Income-growth prediction	Mean baseline wins
3	LGA-season association study	Exposure x disadvantage	Exploratory only; causal and predictive NO-GO
4	Bayesian hierarchy	Y1 transfer to 2021-22	MAE 4.469 vs mean 4.206
5	SIGNAL-F0	Forward Y1 prediction	Mean 4.054 pp; no gain
6	Experiment 5 v1	Mean/ridge/tree/RF	Mean macro MAE 4.054; all models worse
7	E5B zero-control target	Exposed vs explicit-zero FESM	RF 9.4% worse than zero baseline
8	E5C rescue test	Matched controls / flexible RF	Under-training rejected; transfer fails
9	E5D quarterly welfare	Quarterly support-rate outcome	RF loses all 3 forward tests
10	Fiscal Experiment 1	Operating, cash, maintenance	Primary NO-GO
11	Fiscal Experiment 2	Longer matched history	No consistent win over persistence
12	Experiment 2B	Abnormal fiscal shock	Disaster block adds no stable signal
13	2B recovery mechanism	Burden and funding timing	Five-case signal; not robust
14	Experiment 3	Routine maintenance crowd-out	0 certified outcomes; no model
15	Experiment 4 pilot	Budget reallocation	Descriptive GO; causal/model NO-GO
16	Experiment 4 V1	Abnormal budget tree	Tree loses to zero-abnormal baseline
17	Experiment 4B-A	Project completion delay	0/35 complete chains
18	Water utility audit	Disruption vs fiscal capacity	Reshape, then model-training NO-GO
19	Infrastructure acquisition	Three linked sub-experiments	0 causal-ready chains
20	Five-case infrastructure audit	Strongest transport cases	0/5 causal full chains
21	Fitzroy restoration	Daily traffic around reopening	No adjusted rebound
22	Registry common response	Common short/persistent effects	3 short, 2 persistent; no cross-event model
23	Revealed Criticality	F/E/C/S and multimodal ML	Full model loses to mean holdout

I. Bushfire socioeconomic experiments

1. Four-dimension data feasibility audit

Question. Can direct loss, indirect loss, fiscal pressure and social loss be assembled into one community-fire outcome?

Design. Thirty-nine source entries and 48 candidate variables were audited; a bounded NSW prototype linked four fires to six positive LGA-event pairs.

Result. Income and employment trajectories were obtainable. Complete local direct-loss, fiscal-gap and social-loss outcomes were not jointly observed. Release revisions, geographic vintages and temporal leakage prevented a validated four-part target.

GO for bounded outcome-specific construction. NO-GO for the proposed complete monthly four-part index.

2. Six-case bushfire modelling pilot

Question. Can pre-fire income and earner count predict event-year income growth across unseen fires?

Design. Six rows from four fires; leave-one-fire-group-out validation; decision tree and 300-tree random forest against zero and training-mean baselines.

Result. Held-out MAE was 1.100 pp for the mean, 1.269 for the tree and 1.177 for the forest. The models fit the training data but did not transfer.

DATA-CONSTRUCTION GO. SUBSTANTIVE PREDICTION NO-GO.

3. Multi-season LGA association and transferability study

Question. Does disadvantage amplify fire-associated income change, and can a model transfer across seasons?

Design. 3,161 nonduplicated LGA-season candidates; 2,914 Y1 rows across 10 seasons; fixed effects, clustered inference and six forward-held-out seasons.

Result. The main interaction changed sign or collapsed across specifications. Forward MAE was 4.117 pp for the mean and 4.314 for the tree; the tree won 0/6 seasons. The outcome remained nominal resident-income growth, not direct fire loss.

CONDITIONAL GO for exploratory association. CAUSAL AND USEFUL-PREDICTION CLAIMS NO-GO.

4. Bayesian hierarchical trial

Question. Can partial pooling across LGAs and seasons recover transferable signal?

Design. 2,605 training rows, 309 test rows, nine training seasons and a blocked 2021-22 test; Student-t hierarchy with LGA and season random intercepts.

Result. Sampling diagnostics were clean, but Bayesian MAE was 4.469 pp versus 4.206 for the mean. Season variation dominated residual LGA variation. Predictive intervals under-covered.

PREDICTIVE FAILURE. MODEL SOPHISTICATION DID NOT REPAIR TARGET OR TRANSFER LIMITS.

5. SIGNAL-F0 exploratory extension

Question. Can a revised feature design beat the mean on forward-held-out Y1 seasons?

Design. 2,914 rows, 10 seasons and six forward tests under grouped validation.

Result. The best macro MAE was the mean baseline at 4.054 pp. The more complex candidates produced no positive relative gain.

NO-GO FOR CURRENT-DATA PREDICTION; FULL SIGNAL-F REMAINS DATA-BLOCKED.

6. Experiment 5 v1: full model ladder

Question. Do ridge, tree or random forest models improve transfer when the original four-dimensional target is unavailable?

Design. The composite failed its measurement gate. A preregistered reshape used Y1 with six forward season blocks and outcome-window purging.

Result. Macro MAE: mean 4.054, random forest 4.214, tree 4.276, ridge 4.314. Every complex model was worse in all six seasons. Direct loss, fiscal pressure and social impact had zero constructed coverage.

NO DEMONSTRATED TRANSFER ADVANTAGE.

7. E5B: explicit-zero control target

Question. Does centering income growth on explicitly zero-mapped-burn controls reveal a transferable fire signal?

Design. Whole-season tests using retained FESM zero-reference controls and a prespecified random forest comparison.

Result. Random-forest macro MAE was 3.913 pp versus 3.577 for the zero-shock baseline: 9.4% worse, with one win in three tests.

NO DEMONSTRATED TRANSFERABLE SIGNAL.

8. E5C: rescue and under-training diagnostic

Question. Were earlier failures caused by under-training, weak controls or insufficient model flexibility?

Design. Flexible forests, matched-control targets, feature ablation, seed checks and a diagnostic random-row split.

Result. Global-control RF was 9.8% worse than baseline; matched-control RF was 16.4% worse. Training MAE of 0.928 versus held-out MAE of 3.279 showed that the model could fit but did not transfer. Random splits looked better because they mixed seasons and recurring LGAs.

RESCUE FAILED. UNDER-TRAINING HYPOTHESIS REJECTED.

9. E5D: quarterly welfare outcome

Question. Can a more responsive quarterly support-rate outcome reveal transferable exposure and vulnerability signal?

Design. 740 LGA-seasons from six seasons; three forward tests; embargoed prior season; grouped tuning; lower/midpoint/upper confidentiality sensitivities.

Result. Combined RF mean MAE was 8.061 versus 7.518 for the zero baseline. It lost all three forward tests; label-placebo $p=0.710$. The small gain over vulnerability-only RF was 0.051 pp and did not satisfy the full gate.

NO-GO PREDICTIVE SIGNAL.

II. Council finance and reconstruction experiments

10. Fiscal Experiment 1: short-panel decision tree

Question. Can lagged fiscal condition and disaster variables forecast operating ratio, cash cover and maintenance?

Design. Expanding-window future-year validation; 370 primary held-out council-years; fixed shallow tree and simple baselines.

Result. Operating-ratio tree MAE 7.400 versus persistence 6.938. Cash tree MAE 4.317 versus persistence 3.799. Maintenance was a secondary conditional result, but disaster variables appeared in no tree splits.

PRIMARY NO-GO. SECONDARY MAINTENANCE CONDITIONAL ONLY.

11. Fiscal Experiment 2: longer matched history

Question. Does a longer panel make disaster variables predict broad fiscal outcomes consistently?

Design. Eight operating-ratio test years and 818 held-out council-years; fiscal-only, fiscal-plus-disaster and fire specifications; ridge and shallow trees.

Result. No specification met the consistency rule against persistence for operating ratio or cash cover. Maintenance models often beat persistence, but the training mean remained better on MAE and disaster variables added no consistent gain.

NO-GO FOR ESCALATING MODEL COMPLEXITY.

12. Experiment 2B: abnormal fiscal shock

Question. Do disaster variables explain fiscal change beyond each council's fiscal history?

Design. Seven forward-held-out years and 715 operating-ratio predictions. M0 used fiscal history; M1 added current-vintage hazards; M2 required fiscal-transmission evidence.

Result. Operating-ratio ridge MAE was 6.078 for M0, 6.091 for M1 and 6.077 for zero-shock persistence. M2 was too sparse and potentially endogenous to score. Similar instability held across cash and maintenance outcomes.

NO-GO: NO STABLE, AUDITABLE DISASTER-SPECIFIC SIGNAL.

13. Experiment 2B recovery-mechanism pilot

Question. Can gross burden, commitments, approvals and cash timing explain post-disaster fiscal movement?

Design. Seven council-event cases, five core cases; leave-one-council-event-out ridge and depth-two tree; monetary concepts retained separately.

Result. Across all seven cases, zero-change MAE 6.425 beat M2 ridge at 10.201. In the five higher-quality cases, M2 ridge achieved 5.805, 0.529 better than M0, but the gain vanished when reserve cases returned and did not reproduce for maintenance.

CONDITIONAL GO FOR MECHANISM DOCUMENTATION; PREDICTIVE CLAIM NO-GO.

14. Experiment 3: routine maintenance crowd-out

Question. Do disasters displace planned routine maintenance, especially in fiscally constrained councils?

Design. Three affected and three comparison councils; 36 council-years; 37 documentary observations; pre-event comparison screening.

Result. No certified same-year, same-scope routine budget/actual outcome was recovered. Capital disruption and reprioritisation were documented, but routine crowd-out magnitude and fiscal-capacity interaction remained unidentified.

NO-GO FOR CAUSAL ESTIMATION. NO MODEL FITTED.

15. Experiment 4: abnormal budget reallocation pilot

Question. Do disasters cause abnormal downward revisions to pre-planned ordinary infrastructure spending?

Design. Five council-years; 79 budget chains, 312 stage values, 32 selected project records and five annual cash-IPPE pairs.

Result. Large downward revisions existed in treated and comparison councils, but source scopes differed, same-scope final actuals were absent, pretrends were incomplete and asset damage could not be separated from displacement. The model-ready ordinary panel contained zero rows.

DESCRIPTIVE DOCUMENTARY GO. CAUSAL AND MODELLING NO-GO.

16. Experiment 4 reconciliation and V1 shallow tree

Question. Can the expanded quarterly evidence support an abnormal-revision predictor?

Design. The 62-project, 375-stage store was reconciled. A strict V1 retained nine eligible transitions across four councils, two years and one linked episode; whole-council and temporal validation were used.

Result. The fully pruned one-leaf tree had temporal holdout MAE about \$26.8m versus \$18.6m for the zero-abnormal baseline. The broader reconciled store still had zero certified project outcomes.

NEGATIVE PREDICTIVE RESULT; DO NOT INCREASE COMPLEXITY.

17. Experiment 4B-A: project completion delay

Question. Can project-level disaster delay be reconstructed before causal or ML work?

Design. Frozen Northern Rivers cohort of 35 projects; required stable identity, pre-event schedule, event damage/access, revisions and completion or censoring.

Result. Zero of 35 projects completed the required chain. Some month-level completion, contract and deferral evidence was recovered, but identity and schedule continuity remained binding gaps.

0/35 MEASUREMENT GATE. NO COMPARISON COHORT OR MODEL.

Related access studies

Funding-access and betterment-access branches established that approvals, commitments, receipts, gross expenditure and betterment awards must remain separate. They improved the data contract but did not supply an independent predictive experiment or a complete denominator for policy access.

III. Water-service experiments

18A. Water utility feasibility and reshape audit

Question. Does pre-disaster fiscal capacity reduce drinking-water disruption after disasters?

Design. Public utility performance, council finance, fire severity and event reports were audited. The annual panel included hundreds of utility-years, but annual interruption measures were not event-specific restoration outcomes.

Result. Zero rigorously certified event-specific outage observations were obtained. Utility-years, raster pixels and multiple towns could not be treated as independent disasters. A narrower Black Summer association was considered but did not answer the original claim.

RESHAPE FOR VALIDATION ONLY. ORIGINAL CAUSAL CLAIM NO-GO.

18B. Event-level water outcome audit

Question. Did targeted public reports produce a model-ready physical interruption endpoint?

Design. Nineteen candidate dossiers across 14 utilities; 1,620 outcome-source rows, 128 source records and targeted review of reports and notices.

Result. No dossier became a complete utility-event duration observation or a scoped whole-event zero. The certified master outcome/finance/exposure intersection remained empty.

MODEL-TRAINING NO-GO. ABSENCE OF A CERTIFIED ENDPOINT IS NOT A ZERO EFFECT.

IV. Revealed infrastructure criticality experiments

19. National/NSW event-database acquisition experiment

Question. Can physical failure, functional loss, economic activity, restoration and conventional value be linked event by event?

Design. Transport-first registry spanning road, rail, electricity, telecommunications and water; hazard maps, service dates, traffic, businesses and network data were integrated.

Result. The first release catalogued 53 candidates, 24 source-supported disruptions and eight timed transitions, but no event had a quantified network-function change plus an independent economic outcome and comparison. Conventional criticality comparison was not estimable.

CONTINUE MEASUREMENT INITIALLY; DO NOT FIT A FINAL MODEL.

20. Five-case strongest-event completion audit

Question. Do the strongest public cases complete the full disruption-to-economic-recovery chain?

Design. Alice River Bridge, Fitzroy River Bridge, Captain Cook Highway, Palmerston Highway and Naughtons Gap were audited end to end.

Result. Two reached descriptive local traffic chains, two had endpoint proxies and one lacked a post-restoration outcome. Zero of five had a causal full chain.

STOP DATABASE EXPANSION; PIVOT TO FUNCTIONAL/ACTIVITY RESPONSE.

21. Fitzroy Bridge restoration-response model

Question. Did permanent reopening on 10 December 2023 produce a discontinuity in daily traffic?

Design. Daily all-vehicle and heavy-vehicle interrupted time series, HAC uncertainty, 28/56/84-day windows, placebo dates and a same-weekday historical control.

Result. Primary adjusted changes were -15.1% for all traffic and -43.6% for heavy traffic, both uncertain. Seasonal-control estimates were -0.6% and -3.9%, with $p=0.932$ and 0.774 . Staged access and holiday cycles explained the raw fall.

NO EVIDENCE OF A DISTINCT PERMANENT-OPENING REBOUND.

22. Registry-wide common response protocol

Question. Do failures and reopenings produce common short-run and persistent activity responses?

Design. All 54 events retained; identical event-time protocol, HAC uncertainty, bandwidth sensitivity and placebos; missing estimates remained missing.

Result. Three short-run and two persistent responses were estimable. The stable finding was a 26.0% South Australian electricity-demand decline after the 2016 blackout. Bells and Fitzroy restoration responses were bandwidth-sensitive or confounded. Only three event targets existed, so feature-block modelling was not estimable.

RESPONSE SYSTEM WORKS; COMMON REBOUND AND CROSS-EVENT PREDICTION NOT SUPPORTED.

23. Revealed Infrastructure Criticality

Question. Do observed multidimensional consequences differ from conventional criticality, and can multimodal features predict them across unseen disasters?

Design. F/E/C/S fingerprints for 54 events; development-only standardisation; 20,000 plausible weight draws; ELLIE_2023 and JASPER_2023 final holdouts; whole-family Ridge comparisons and block ablation.

Result. Coverage was F=13, E=9, C=2, S=7; only two events had at least three dimensions and one had all four. Four events supported a descriptive conventional comparison. On the 13-event functional target, holdout MAE was 0.602 for the mean versus 0.750 hazard, 0.679 conventional, 0.639 network-combined and 0.631 full multimodal.

MEASUREMENT SYSTEM COMPLETE. SCALAR, HIDDEN-CRITICALITY AND MULTIMODAL PREDICTION CLAIMS NOT ESTABLISHED.

Cross-experiment synthesis

What repeatedly failed

Transferability. Models that fit historical councils, LGAs or events did not reliably improve future-season or unseen-disaster error.

Outcome validity. Annual income and fiscal ratios were observable but indirect; direct loss, service restoration, fiscal transmission and social burden rarely overlapped at the same event unit.

Independence. Thousands of rows often represented repeated LGAs or shared disaster seasons. The effective number of independent shocks was small.

Comparability. Budgets, actuals, grants, reconstruction costs, traffic, customer counts and network proxies used different scopes and clocks.

Complexity. Bayesian hierarchies, forests and expanded feature blocks did not rescue weak targets or temporal distribution shift.

What is defensible to tell Bowen

AUSSEF has not found a validated predictive model of disaster socioeconomic impact or infrastructure criticality. It has built a reproducible measurement framework and shown, through repeated grouped and forward tests, why apparent patterns from random splits, aggregate annual outcomes or incomplete event chains do not generalise. The 2016 South Australian blackout supplies one stable functional response; the remaining positive signals are descriptive or fragile sensitivities.

Current research decision

STOP MODEL SEARCH ON THE PRESENT TARGETS. RETAIN THE DATABASE, PROTOCOLS AND NEGATIVE RESULTS.

A new positive claim would require a genuinely new outcome or new independent events, specified before modelling. Reweighting the same proxies, adding algorithms or selecting the most favourable branch would not create new evidence.

Terminology

NO-GO means the tested specification did not meet its evidence gate. It does not mean disasters have no effect.

DATA-BLOCKED means the target could not be measured at the required unit. CONDITIONAL GO means a bounded descriptive or measurement study remains useful, without supporting the stronger causal or predictive claim.

Source record

This summary consolidates the final or controlling result documents listed below. Duplicate mirrors and exact reruns were treated as one experiment; distinct targets, validation designs and decision gates were retained separately.

Controlling artifact
aussef_pilot_input/deliverables/DATA_FEASIBILITY_REPORT.md
aussef_pilot/MODEL_RESULTS.md
bushfire_lga_season/outputs/FINAL_RESEARCH_VERDICT.md
bayesian_hierarchical_trial/RESULTS.md
signal_f_experiment/outputs/RESULTS.md
experiment_5/results/EXPERIMENT_5_EXECUTION_REPORT.md
experiment_5/e5b/results/E5B_EXECUTION_REPORT.md
experiment_5/e5c/results/E5C_EXECUTION_REPORT.md
experiment_5/e5d/results/E5D_EXECUTION_REPORT.md
/Users/ray/Research/AUSSEF/Experiment 1/results/interpretation.md
/Users/ray/Research/AUSSEF/Experiment 2/results/README.md
experiment_2b/EXPERIMENT_2B_RESULTS.md
experiment_2b/EXPERIMENT_2B_RECOVERY_MODEL_RESULTS.md
/Users/ray/Research/AUSSEF/Experiment 3/results/measurement_quality.md
/Users/ray/Research/AUSSEF/Experiment 4/results/PILOT_REPORT.md
experiment4_abnormal_budget_reallocation/RECONCILIATION_REPORT.md
/Users/ray/Research/AUSSEF/Experiment 4/modelling_v1/results/final_results.json
experiment_4b_a/EXPERIMENT_4B_A_RESULTS.md
audit_water/FEASIBILITY_AUDIT.md
water_research/FEASIBILITY_REPORT.md
infrastructure_event_lab/reports/EXPERIMENT_RESULTS.md
infrastructure_event_lab/reports/SHORTLIST_COMPLETION.md
infrastructure_event_lab/reports/FITZROY_RESTORATION_MODEL.md
infrastructure_event_lab/reports/REGISTRY_RESPONSE_RESULTS.md
infrastructure_event_lab/reports/REVEALED_CRITICALITY_RESULTS.md

Prepared from the locally retained AUSSEF evidence base. Numerical statements remain bounded by the definitions and caveats in their controlling artifacts.